

COUNTING LATTICE POINTS THE HARD WAY

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ABSTRACT. This is a fun addendum to the author's previous paper [1].

Definition 0.1. The *Leech lattice* is the unique 24-dimensional even unimodular lattice in 24 dimensions such that all nonzero vectors have norm at least 2.

Question 0.2. How many vectors in the Leech lattice have norm 2?

We could answer this question simply by picking basis vectors and starting to count. However, there's a faster way. Recall from [1] that every even unimodular lattice gives rise to a *lattice VOA* whose *character* is given by

$$(0.3) \quad q^{-\text{rank}(\Lambda)/24} \sum_n \dim V_n q^n = \frac{\Theta_\Lambda}{\eta^{\text{rank}(\Lambda)}}$$

In the case of lattice VOAs specifically, we saw that this character must necessarily be of the form $\alpha j + \beta$, where j is the *j-invariant* and α and β are coefficients. We can determine what the coefficients are, and therefore what the character must be, simply by computing the first two terms. Recall that the *degree* of an element of a lattice VOA is given in two parts.

$$(0.4) \quad \deg(m|\lambda\rangle) = \deg(m) + \frac{1}{2}|\lambda|^2$$

From this, we see that there exists exactly one element of degree 0: the vacuum element $|0\rangle$.

Now consider elements of degree 1. Since all nonzero vectors have norm at least 2, we know that all such elements must arise from creation operators b_{-1}^n acting on the vacuum element. There are 24 such creation operators, so there are 24 elements of degree 1. This means that

$$(0.5) \quad \alpha j + \beta = q^{-1} + 24 + \dots$$

so the character must be $j - 720$. Finally, we can compute the number of degree 2 elements two different ways. First, using creation operators we can compute

element	number
$ \lambda\rangle, \lambda ^2 = 4$	unknown
$b_{-2}^n 0\rangle$	24
$b_{-1}^n b_{-1}^n 0\rangle$	24
$b_{-1}^i b_{-1}^j 0\rangle, i \neq j$	$\binom{24}{2}$

noting that the b_{-1}^i all commute with each other. Observing next from the character that there are 196884 elements of degree 2, we know that the number of lattice points with norm 2 is

$$(0.6) \quad 196884 - \binom{24}{2} - 24 - 24 = 196560$$

This number is notable because the Leech lattice gives the optimal packing of 24-dimensional spheres, so 196560 is the *kissing number* in 24 dimensions.

REFERENCES

- [1] R. O'Farrell and D. Saunders, *An Introduction to Monstrous Moonshine*.
<https://ro-farrell.github.io/writeups/moonshine.pdf>